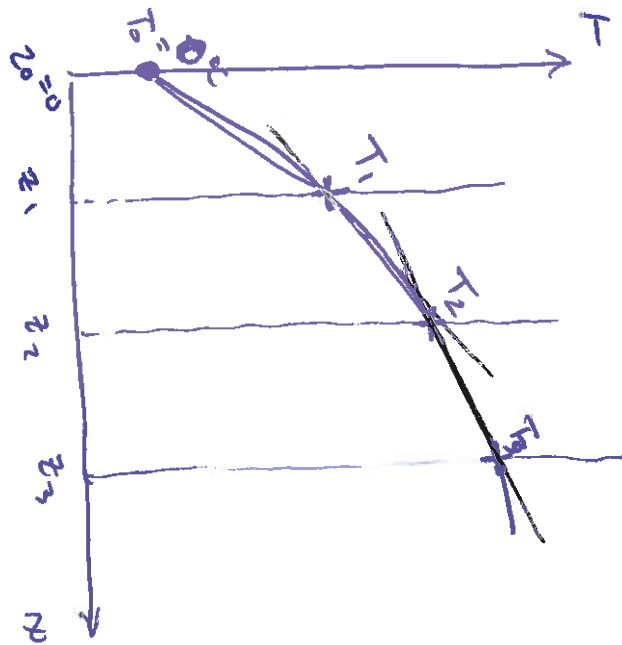
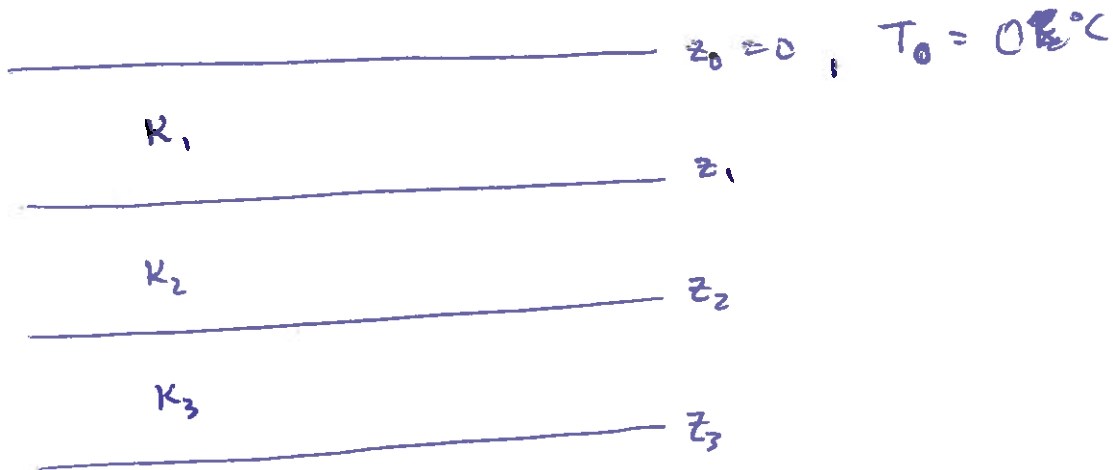


$$\frac{\partial T}{\partial t} = K \nabla^2 T$$

HEAT EQUATION

$$1-D \quad \frac{\partial T}{\partial t} = K \frac{\partial^2 T}{\partial z^2}$$



$$\frac{\partial T}{\partial t} = K \frac{\partial^2 T}{\partial z^2}$$

$$\text{1st DERIVATIVE} \quad \frac{\partial T(z_2)}{\partial z} \approx \frac{T_2 - T_1}{z_2 - z_1}$$

BACKWARDS DIFFERENCE
 $z_2 - z_1 = \Delta z$

$$\approx \frac{T_3 - T_2}{z_3 - z_2}$$

FORWARD DIFFERENCE
 $z_3 - z_2 = \Delta z$

CENTRAL DIFFERENCE (AVERAGE OF FORWARD & BACK)

$$\frac{1}{2} \text{ BACKWARDS DIFF} = \frac{T_1 - T_2}{z_1 - z_2}$$

$$\frac{1}{2} \text{ FORWARD} + \frac{1}{2} \text{ BACKWARD} = \text{CENTRAL}$$

$$\frac{1}{2} \left(\frac{T_3 - T_2}{\Delta z} \right) + \frac{1}{2} \left(\frac{T_1 - T_2}{\Delta z} \right) =$$

$$\frac{T_3 - 2T_2 + T_1}{2\Delta z} \approx \frac{\partial T(z_2)}{\partial z}$$

FOR ANY NODE, i

$$\frac{T_{i+1} - 2T_i + T_{i-1}}{2\Delta z} \approx \frac{\partial T(z_i)}{\partial z}$$

2ND ORDER DERIVATIVE (CENTRAL DIFF APPROX.)

$$\frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta z^2} \approx \frac{\partial^2 T(z_i)}{\partial z^2}$$

$$\frac{\partial T(t_j)}{\partial t} \approx \frac{T^{j+1} - T^j}{\Delta t}$$

EXPLICIT (LOOKING FORWARD IN TIME)

$$\frac{\partial T(t_j)}{\partial t} \approx \frac{T^j - T^{j-1}}{\Delta t}$$

IMPLICIT (LOOKING BACKWARD IN TIME)

(3)

EXPLICIT

$$\frac{\partial T}{\partial t} = K \frac{\partial^2 T}{\partial z^2}$$

$$\frac{T_i^{j+1} - T_i^j}{\Delta t} = K \frac{T_{i+1}^j - 2T_i^j + T_{i-1}^j}{\Delta z^2}$$

$$T_i^{j+1} = T_i^j + \frac{K\Delta t}{\Delta z^2} (T_{i+1}^j - 2T_i^j + T_{i-1}^j)$$

STABILITY CONDITION

$$K \frac{\Delta t}{\Delta z^2} < 0.5$$

IMPLICIT

$$\frac{T_i^{j+1} - T_i^j}{\Delta t} = K \frac{T_{i+1}^{j+1} - 2T_i^{j+1} + T_{i-1}^{j+1}}{\Delta z^2}$$

$$-K \frac{\Delta t}{\Delta z^2} T_{i-1}^{j+1} + \left(1 + 2K \frac{\Delta t}{\Delta z^2}\right) T_i^{j+1} - K \frac{\Delta t}{\Delta z^2} T_{i+1}^{j+1} = T_i^j$$

$$K \frac{\Delta t}{\Delta z^2} = \alpha, \quad -\alpha T_{i-1}^{j+1} + (1 + 2\alpha) T_i^{j+1} - \alpha T_{i+1}^{j+1} = T_i^j$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\alpha & (1+2\alpha) & -\alpha & 0 & & & \\ 0 & -\alpha & (1+2\alpha) & -\alpha & 0 & & \\ & & \vdots & & & & \\ & & -\alpha & (1+2\alpha) & -\alpha & & \\ & & 0 & & & 1 & \end{bmatrix} \begin{bmatrix} T_0^{j+1} \\ T_1^{j+1} \\ T_2^{j+1} \\ \vdots \\ T_{n-1}^{j+1} \\ T_n^{j+1} \end{bmatrix} = \begin{bmatrix} T_0^j \\ T_1^j \\ T_2^j \\ \vdots \\ T_{n-1}^j \\ T_n^j \end{bmatrix}$$

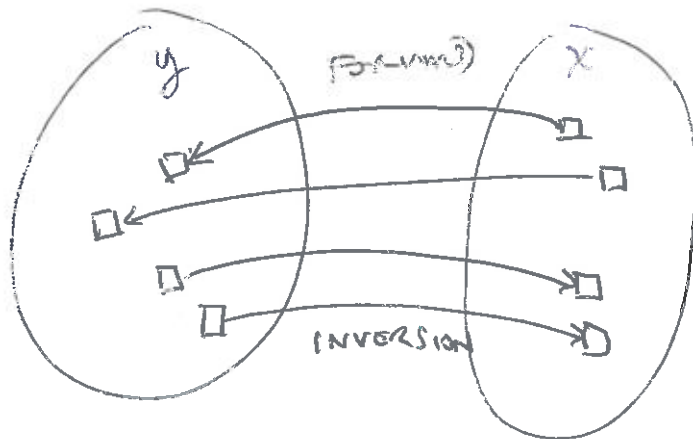
FORWARD PROBLEM

$$g_z = \int \frac{\rho(x, y, z)}{r} dr$$

$$y = mx + b$$

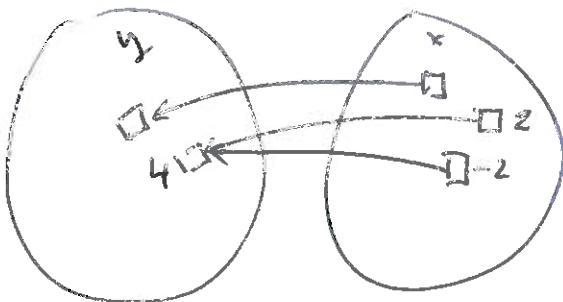
SIMPLE INVERSION :

$$\frac{y - b}{m} = x$$



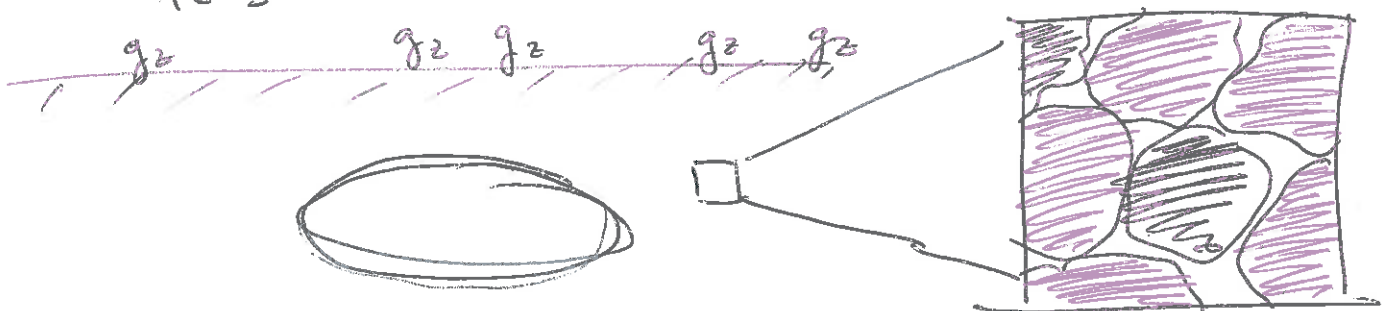
$$y = x^2 \text{ FORWARD}$$

$$x = y^{1/2} \text{ INVERSION}$$



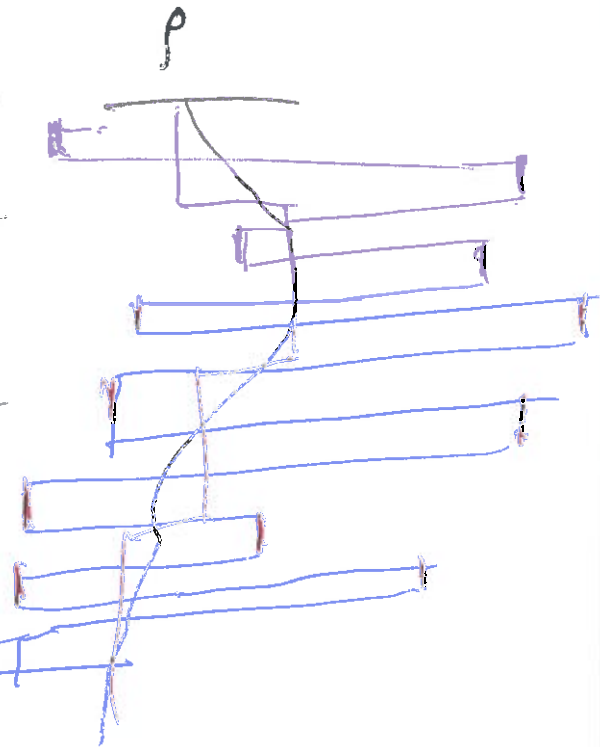
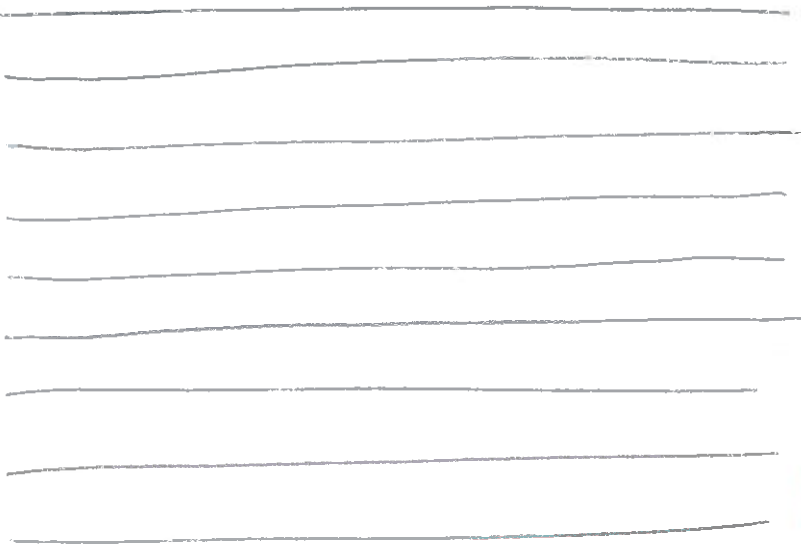
NON-UNIQUENESS
MULTIPLE SOLN FOR
SINGLE VALUES

UNDER-DETERMINED → OVER DETERMINED
NOT ENOUGH DATA TO SOLVE SIMPLY



STABILITY

z ↓



3 ISSUES

NONUNIQUENESS

UNDERDETERMINED

UNSTABLE

LINEAR LEAST SQUARES
mathematical model

$$\tilde{y}_1 = mx_1 + b$$

$$\tilde{y}_2 = mx_2 + b$$

$$\tilde{y}_3 = mx_3 + b$$

$$\tilde{y}_N = mx_N + b \rightarrow \text{model (unknown)}$$

data (known)

$$y_1, x_1$$

$$y_2, x_2$$

$$y_3, x_3$$

$$y_N, x_N$$

↑
pseudo data (modeled data)
↑
position (known)

(6)

MISFIT BETWEEN DATA & MODEL

$$\phi_i = y_i - \tilde{y}_i$$

$$\phi^2 = \sum_{i=1}^N (y_i - \tilde{y}_i)^2$$



WANT TO MINIMIZE MISFIT

DERIVATIVE SET TO 0

$$\begin{aligned} \frac{\partial \phi^2}{\partial m} = 0 &= \frac{\partial}{\partial m} \left(\sum_{i=1}^N (y_i - \tilde{y}_i)^2 \right) \\ &= -2 \sum (y_i - \tilde{y}_i) \frac{\partial \tilde{y}_i}{\partial m} \end{aligned}$$

$$0 = \sum_{i=1}^N y_i x_i - m \sum_{i=1}^N x_i^2 - b \sum_{i=1}^N x_i$$

$$\frac{\partial \phi^2}{\partial b} = 0 = \sum_{i=1}^N y_i - m \sum_{i=1}^N x_i - b N$$

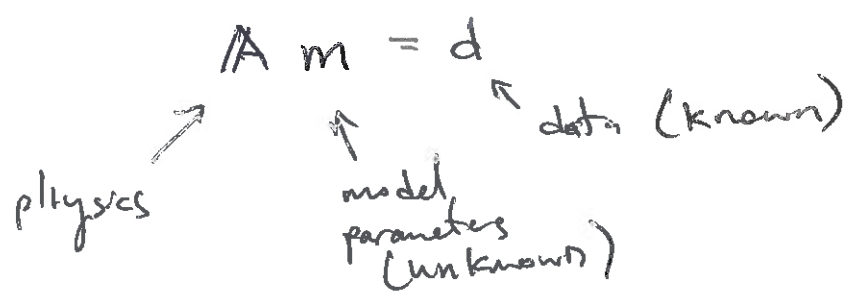
$$m = \frac{N \sum_{i=1}^N y_i x_i - \sum_{i=1}^N y_i \sum_{i=1}^N x_i}{N \sum_{i=1}^N x_i^2 - \left(\sum_{i=1}^N x_i \right)^2}$$

$$b = \frac{\sum_{i=1}^N y_i \sum_{i=1}^N x_i^2 - \sum_{i=1}^N y_i x_i \sum_{i=1}^N x_i}{\sum_{i=1}^N x_i^2 - \left(\sum_{i=1}^N x_i \right)^2}$$



$$\begin{aligned}
 y_i \\
 b + mx_1 &= \tilde{y}_1 \\
 b + mx_2 &= \tilde{y}_2 \\
 b + mx_3 &= \tilde{y}_3 \\
 &\vdots \\
 b + mx_N &= \tilde{y}_N
 \end{aligned}$$

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ \vdots & \vdots \\ 1 & x_N \end{bmatrix} \begin{bmatrix} b \\ m \end{bmatrix} = \begin{bmatrix} \tilde{y}_1 \\ \tilde{y}_2 \\ \tilde{y}_3 \\ \vdots \\ \tilde{y}_N \end{bmatrix}$$



LINEAR LEAST SQUARES

$m = A^{-1}d$ only possible if A is a square matrix

$$Am = d$$

$$\underbrace{(A^T A)}_{\text{square}} m = \underbrace{A^T}_{\text{transpose}} d$$

$$m = (A^T A)^{-1} A^T d$$

MATLAB
 $m = \text{inv}(A^T A) * A^T * d$
 $m = A \backslash d$

$$\tilde{d} = \frac{1}{A} m$$

⑧

$$\phi^* = \|d - \tilde{d}\|_2$$